Transport of a morphism along identities computes as conjugation by idtoiso

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:idtoiso-trans' (source lines 167-173, Lemma 9.1.9); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The displayed equation carries the internal label ct:idtoisocompute, which is defined and used inside the retained statement and proof, so it needs no external entry.

Editorial note. The retained counter values (chapter 9, section 1, theorem offset 8) make the counted lemma print as Lemma 9.1.9, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: the computation of transport along identities in a precategory (Lemma 9.1.9).

Allowed standard prerequisites (editorial).

- idtoiso. For p : a = a' the isomorphism idtoiso(p) : a ~ a' is defined by induction from the identity isomorphism.. It is the conjugation map appearing on the right-hand side of the displayed equation.
- Transport in a hom-type. For p : a = a' and q : b = b' the transport of f : hom_A(a,b) along (p,q) is a morphism hom_A(a',b').. It is the left-hand side of the displayed equation.
- Path induction. To prove a statement about paths it suffices to prove it for reflexivity.. It is the sole proof step at source line 175.

Lemma 9.1.9. *For $p : a = a'$ and $q : b = b'$ and $f : \hom_A(a, b)$, we have*

$$(p, q)_*(f) = \mathsf{idtoiso}(q) \circ f \circ \mathsf{idtoiso}(p)^{-1}. \tag{9.1.10}$$

Proof. By induction, we may assume p and q are refl_a and refl_b respectively. Then the left-hand side of (9.1.10) is simply f. But by definition, $\mathsf{idtoiso}(\mathsf{refl}_a)$ is 1_a, and $\mathsf{idtoiso}(\mathsf{refl}_b)$ is 1_b, so the right-hand side of (9.1.10) is $1_b \circ f \circ 1_a$, which is equal to f. □